IV Infusions Practice 2

Module 3 - Intravenous Infusions · Daily practice sheet · No date

Name _____ Date _____

An antibiotic drug is to be given to an adult male patient (162 lb, 58 years old) by IV infusion. The drug is supplied in sterile vials containing 30 mL of the antibiotic solution at a concentration of 125 mg/mL. Assume the drug follows the pharmacokinetics of a one-compartment open model. The apparent volume of distribution of this drug is 0.5 L/kg, and the elimination half-life is 3.1 hours.

a What rate in milliliters per hour would you infuse this this drug to obtain a steady-state concentration of 20 mcg/mL in this patient?

b What loading dose would you suggest?

c How many milliliters of the antibiotic solution are required for the loading dose?

d What is the expected concentration of drug in the plasma at the end of a 6 hour infusion without a loading dose?

e What is the concentration of drug in the plasma 12 hours after the cessation of the 6-hour infusion (no loading dose)?

f If an appropriate loading dose were administered with the simultaneous IV infusion, how much time after the cessation of the infusion would be required for the concentration to reach 5 mcg/mL?